Tool-Error-Induced Next-Action Direction in Agentic LLMs

Anonymous Author(s)

Abstract

We report a pre-registered mechanistic interpretability study of next-action choice after tool failure across three 8B-class language models spanning agent-training depth: Qwen3-8B, Hermes-3-Llama-3.1-8B, and Llama-xLAM-2-8b-fc-r. A linear probe trained on the residual stream at the post-tool-result token (position t^*) achieves **90.0%, 86.0%, and 90.1%** balanced accuracy on the 3-class {retry-same, modify-args, give-up} decision (chance 33%; balanced shuffle floor 33.8%, 33.6%, 33.5%), with permutation $p < 10^{-4}$ surviving Holm-Bonferroni correction. The probe outperforms a position-only baseline by 27–33 pp. Causal steering at α=2.5 converts give-up trajectories to any non-give-up action in **100%** (Qwen3-8B; n=60, file_read-dominated—see §**??**), **92.9%**, and **99.0%** of cases vs. 5.0%, 5.5%, and 25.1% for random-direction controls (Cohen's $h > 1.86$ for all three models). Critically, a *subspace-matched* random control—a random vector confined to the action subspace span$(v_{\text{retry}}, v_{\text{modify}}, v_{\text{give-up}})$ at the same norm—achieves only 33.3%, 32.5%, and 40.2% broad flip, confirming that the *specific orientation* of v_{retry} within the action subspace matters, not merely the norm or the subspace energy. The *form* of recovery varies with training depth: Qwen3-8B produces exclusively modify-args (0% retry-same); Hermes-3 mixes both (44.7% retry-same); xLAM-2 favours retry-same (56.7%). This gradient is sharpest on the calculator tool, where the retry-same rate under steering is 6.3% (Hermes-3) vs. 34.9% (xLAM-2), while both models achieve ≈100% broad flip. A structural finding: the leave-one-tool-out probe fails for calculator (27–44% in Hermes-3 and xLAM-2, the two models with calculator give-ups) yet the causal direction still forces continuation at near-100%—a probe-vs-causal dissociation geometrically explained by cross-tool CKA ≈ 0.01 (random null ≈ 0.14; geometry characterised in Hermes-3 and xLAM-2). The research loop was driven by the same LLM agent that is the study's subject; the executor and subject models are never the same model in any given run, and probes and interventions are deterministic functions of cached activations, so the executor's policy cannot influence the measurements.

1 Introduction

Agentic language models make tool calls; tools fail; the model must decide to retry, modify its arguments, or give up. This choice is consequential for agentic reliability, yet its internal mechanism has not been characterised at the residual-stream level.

Related work. Healy et al. (2026) show that internal representations during a single forward pass predict tool-selection hallucinations at up to 86.4% accuracy, targeting *detection* of wrong tool choices before calling; our work targets *causal steering* of recovery choices *after* a returned failure—a complementary operation. Yin et al. (2025) independently locate relevant geometry in late residual streams: their finding that RL reasoning enhancement "disproportionately collapses tool-reliability-related representations, and hallucinations surface as amplified divergences concentrated in late-layer residual streams" is consistent with our probes peaking at layers 28–33 (of 32–36 total). Cheng et al. (2025) introduce FASB, a dynamic activation-steering framework for alignment tasks; our work complements this by studying one specific behavioral dimension (tool-error recovery) under pre-registered experimental controls.

Methodological foundations. Zou et al. (2023) establish that mean-difference directions in residual streams are a principled basis for linear control vectors. Li et al. (2023) demonstrate that such directions

causally modulate factual accuracy when added back at inference time, which is the direct precedent for our α-sweep intervention. Todd et al. (2024) show that compact linear directions in transformer residual streams encode executable in-context functions and can transfer across contexts. Heimersheim and Nanda (2024) provide methodological grounding for causal intervention design including null comparisons. Heo et al. (2024) find an instruction-following dimension that generalises across unseen tasks but not across unseen instruction types—a direct empirical parallel to our LOO finding (§4). For representational geometry we use Centered Kernel Alignment (Kornblith et al., 2019), surveyed by Klabunde et al. (2023).

Contributions. (1) A pre-registered probe-and-steer experiment across three models with a full sanity-gate suite. (2) A *universal* probe-vs-causal dissociation for the calculator tool, explained by cross-tool identity orthogonality together with a give-up→continue action axis that is shared between the transferring tools (web_search↔file_read, cos ≈ 0.6) but orthogonal for calculator. (3) A training-depth gradient in recovery richness—the narrow retry-same rate under steering correlates with agent-specific training density across all models.

2 Methods

Stimuli. 300 base agentic prompts (100 per tool $\in$ {web_search, calculator, file_read}), each paired with 4 outcome conditions (success, rate_limited, no_results, invalid_arg). Mock tool responses are byte-deterministic via SHA-256(tool $\|$ args $\|$ condition_seed). Three seeds; splits disjoint by base prompt: 180 probe-fit / 60 direction-extract / 60 causal-test. **Disclosure:** some conditions (notably no_results) return seed-invariant content; multi-seed results reflect repeated measurement for those conditions, not independent replication.

Subjects. **Qwen3-8B** (36 layers, d=4096, thinking mode disabled): deep agentic post-training, BFCL v3 scored. Produces very few give-up actions ($\approx$2% of error trajectories); only n=60 give-up cases reach the causal test set, of which n=57 are file_read, n=3 web_search, and n=0 calculator. **Hermes-3-Llama-3.1-8B** (32 layers): SFT on `hermes-function-calling-v1` ($\approx$10k examples); n=255 give-up cases. **Llama-xLAM-2-8b-fc-r** [G4] (32 layers): trained on `xlam-function-calling-60k` + `APIGen-MT-5k` (multi-turn agentic examples from simulated agent-human interactions, both public on HuggingFace); n=291 give-up cases. Tool-call format: bare JSON array.

Labelling. Strict regex parser, first-match: (1) *retry-same*—same tool call, token-identical arguments; (2) *modify-args*—changed arguments or different tool; (3) *give-up*—no tool call within 256 tokens plus final answer, "I cannot…", or end-of-turn; (4) *ambiguous* (excluded). Ambiguous rate: 0% in all models.

Probe. 3-way ℓ_2-regularised logistic regression per layer, restricted to error trajectories with action-balanced downsampling. Class directions follow Zou et al. (2023): $\mathbf{v}_c = \text{mean}(\mathbf{h} \mid y{=}c) - \text{mean}(\mathbf{h} \mid y{\neq}c)$, computed on the direction-extract split.

Causal intervention. Norm-matched additive patch at best-layer $\times$ t^* position: $\mathbf{h} \leftarrow \mathbf{h} + \alpha \frac{\mathbf{v}_{\text{retry}}}{\|\mathbf{v}_{\text{retry}}\|} \cdot \|\mathbf{h}\|$, $\alpha \in \{0, 1.0, 2.5, 5.0\}$. Null controls: random Gaussian direction, orthogonal projection, Gaussian noise (Heimersheim and Nanda, 2024). Two metrics: *narrow* (give-up → retry-same) and *broad* (give-up → any non-give-up). Broad is primary for G4 (preregistered after G3 showed Qwen3-8B never produces retry-same).

Statistics. Permutation test for probe accuracy (10k shuffles; Holm-Bonferroni, family-wise α=0.05). Cluster bootstrap by base-prompt for flip rates (10k resamples; percentile bootstrap CI). Cohen's h effect size. Cross-family replication is independent (no pooling).

Table 1: Probe accuracy at best layer. Chance = 33%. All permutation tests: $p < 10^{-4}$.

Model	Best layer	Test acc	Holm-surviving	PC1 var.
Qwen3-8B	33 / 36	**89.4%**	36 / 36	86.9%
Hermes-3	28 / 32	**80.6%**	32 / 32	90.6%
xLAM-2	29 / 32	**89.4%**	31 / 32	68.4%



Figure 1: Per-layer 3-way probe accuracy for all three models (error-only, action-balanced). Stars mark best-probe layers (L33, L28, L29). All models exceed the 60% threshold from layer 20 onward.

CKA. Linear CKA (Kornblith et al., 2019) via kernel ($n \times n$) Gram matrices in float64. Raw CKA is undefined when $d \gg n$ ($\mathbb{E}[\text{CKA}(\mathbf{X}_{\text{rand}}, \mathbf{Y}_{\text{rand}})] \approx d/(d+n) \approx 0.93$ for d=4096, n=300); we apply PCA-50 reduction first, yielding empirical null $\approx$0.143 for n=300 and $\approx$0.077 for n=600 (theoretical $k/(k+n)$ agrees to <0.001). Sanity suite: CKA(X, X)=1.0, symmetry, rotation invariance, primal≡kernel cross-check, null>0.6 for raw formula, null<0.50 for PCA version—all passed.

3 Results

3.1 Linear Probes (H1a)

All three models clear the pre-registered 60% threshold by a large margin, with permutation $p < 10^{-4}$ (Table 1). Best layers lie in the top 8–11% of each network (L33/36, L28/32, L29/32), consistent with the late-layer tool-reliability geometry reported by Yin et al. (2025). PC1 of the three class-mean directions explains 86.9%, 90.6%, and 68.4% of class-separation variance (gate G-PCA ≥40%, all pass), confirming a dominant one-dimensional separation geometry (Todd et al., 2024).

Within-error-type probe accuracy is consistent across all models and error types (83–98%), confirming the direction is not error-type-specific. The probe also substantially outperforms a position-only baseline (89.4% vs. 62.2% for Qwen; 80.6% vs. 50.0% for Hermes-3; 89.4% vs. 56.1% for xLAM-2), establishing that residual-stream *content* encodes action-choice information beyond trajectory structure. Because the held-out test set is class-imbalanced (majority-class fractions 62/47/56%), the raw test accuracies in Table 1 are the per-layer

Table 2: Broad flip rate (give-up → any non-give-up), α=2.5. Subspace-rand: random vector in span(v_r, v_m, v_g), norm-matched. Subscripts are Wilson 95% CIs.

Model	Retry-dir	Subspace-rand	Random-dir	$\Delta_{\text{ret-rand}}$ (pp)	Cohen's h
Qwen3-8B	$100.0\%_{[94,100]}$	$33.3\%_{[23,46]}$	5.0%	**95.0**	2.67
Hermes-3	$92.9\%_{[89,96]}$	$32.5\%_{[27,39]}$	5.5%	**87.4**	2.15
xLAM-2	$99.0\%_{[97,100]}$	$40.2\%_{[35,46]}$	25.1%	**73.9**	1.89

Table 3: Narrow flip rate (give-up → retry-same exactly), α=2.5. Subscripts are Wilson 95% CIs.

Model	Retry-dir	Random-dir	Δ (pp)	Cohen's h
Qwen3-8B	$0.0\%_{[0,6]}$	0.0%	0.0	0.00
Hermes-3	$44.7\%_{[39,51]}$	2.4%	**42.4**	1.16
xLAM-2	$56.7\%_{[51,62]}$	14.1%	**42.6**	0.94

selection metric, but we report *balanced* accuracy (macro-recall) as the headline figure—**90.0%, 86.0%, and 90.1%**—for which the shuffled-label floor sits at chance (33.8/33.6/33.5%; Appendix A). The two-tier reporting prevents the imbalance from inflating the headline. Leave-one-tool-out transfer and its geometric explanation are discussed in §3.4.

3.2 Causal Intervention (H1b)

Aggregate results (α=2.5). Tables 2 and 3 show the broad and narrow flip rates. H1(b) narrow: Hermes-3 and xLAM-2 clear ($\Delta \geq$25 pp); Qwen3-8B does not (Δ=0 pp). H1(b) broad: all three models clear ($\Delta \geq$72 pp, $h > 1.86$, all models).

All three models maintain $\Delta \geq 59$ pp and $h > 1.86$ against the random-direction null, confirming direction-specificity. The subspace-matched control confirms the specific direction—not merely norm or action-subspace energy—is required (Table 2).

Dose-response. Table 4 shows the broad flip rate across the finer α sweep. The relationship is smooth and monotonic across all three models, saturating at $\alpha \geq 1.5$ for Qwen3-8B and near $\alpha = 2.5$ for Hermes-3.

Per-tool breakdown. Table 5 shows both metrics per tool. The broad metric reveals that all models achieve near-100% flip for *every* tool where give-up trajectories exist—the direction forces any continuation universally. The narrow metric reveals the training-depth gradient most clearly on calculator: Hermes-3 converts only 6.3% of steered calculator give-ups to retry-same (the rest become modify-args); xLAM-2 converts 34.9%. This reflects the density of exact-retry examples in xLAM-2's `APIGen-MT-5k` training data, where transient failures warrant identical retries. Qwen3-8B has zero calculator give-up trajectories in the test set and cannot be evaluated on this tool.

Form of continuation. At α=2.5: Qwen3-8B produces 100% modify-args and 0% retry-same across all 60 give-up cases; Hermes-3 produces 44.7% retry-same, 49.4% modify-args, 5.9% still give-up; xLAM-2 produces 56.7% retry-same, 42.3% modify-args, 1.0% still give-up.

3.3 Cross-Model Replication (xLAM-2)

xLAM-2 independently clears both primary pre-registered thresholds: 90.1% balanced probe accuracy (89.4% raw; $\gg$60% threshold, $p < 10^{-4}$) and 73.9 pp broad flip rate ($\gg$25 pp threshold, h=1.89). This constitutes an independent replication across a third model family with distinct architecture lineage (Llama-3.1) and

Table 4: Broad flip rate vs. steering α (retry direction).

Model	α=0	0.5	1.0	1.5	2.0	2.5
Qwen3-8B	0%	35%	75%	100%	100%	100%
Hermes-3	4%	28%	48%	66%	78%	93%
xLAM-2	0%	38%	70%	85%	93%	99%

Table 5: Per-tool flip rates at α=2.5, retry direction. B = broad; N = narrow. *n=3: statistically unreliable. Wilson 95% CIs shown in brackets.

	Qwen3-8B		Hermes-3		xLAM-2	
Tool	B	N	B	N	B	N
web_search	100%*	0%*	100%	60.5%	100%	63.6%
	n=3		n=114		n=66	
	[44,100]; [0,56]		[97,100]; [51,69]		[94,100]; [52,74]	
calculator	*N/A (n=0)*		100%	6.3%	100%	**34.9%**
			n=48		n=129	
			[93,100]; **[2,17]**		[97,100]; [27,43]	
file_read	100%	0%	83.9%	45.2%	96.9%	81.3%
	n=57		n=93		n=96	
	[94,100]; [0,6]		[75,90]; [35,55]		[91,99]; [72,88]	

agent training corpus (`APIGen-MT-5k`). The key comparative finding is that while all three models share a causally effective continuation direction, the *form* of recovery they express under steering diverges with training depth—xLAM-2 produces retry-same in 56.7% of steered trajectories vs. 44.7% (Hermes-3) and 0% (Qwen3-8B). Cross-tool leave-one-out transfer and its geometric interpretation are reported in the following section.

3.4 Representational Geometry (CKA)

Cross-model similarity. All three models achieve CKA $\approx$ 0.80 at their best action-choice layers on 900 matched trajectories (random null $\approx$ 0.077), 10$\times$ above chance (Table 6). The Hermes-3 vs. xLAM-2 layer-wise profile peaks at L15–16 (CKA $\approx$ 0.953) and drops to its minimum at exactly the best action-choice layers (L28: 0.793, L29: 0.800). Agentic fine-tuning specifically reshapes late-layer representations while early layers remain shared—consistent with the base Llama-3.1-8B shared by Hermes-3 and xLAM-2.

Table 6: Cross-model CKA (PCA-50, n=900; null $\approx$ 0.077 computed at n=600).

Pair	CKA	Ratio to null
Qwen3-8B vs. Hermes-3	0.800	10.4$\times$
Qwen3-8B vs. xLAM-2	**0.874**	11.4$\times$
Hermes-3 vs. xLAM-2	0.798	10.4$\times$

Cross-tool orthogonality and probe transfer. Within each model, pairwise CKA between tool-specific trajectory sets (n=300/tool, null $\approx$ 0.143) ranges from 0.006 to 0.017—near zero across all model $\times$ tool-pair combinations (Figure 3). The three tools therefore occupy essentially orthogonal subspaces at the action-choice layer, a structural property that replicates across all three model families. This orthogonality determines probe transfer: a direction extracted from web_search and file_read trajectories cannot, by construction, generalise to the orthogonal calculator subspace, so LOO probe accuracy for calculator (27–44%) falls below the within-tool accuracy. Conversely, LOO transfer between web_search and file_read (50.0–74.3%) holds precisely because those two tools share subspace alignment. Critically, the causal direction is not subject to

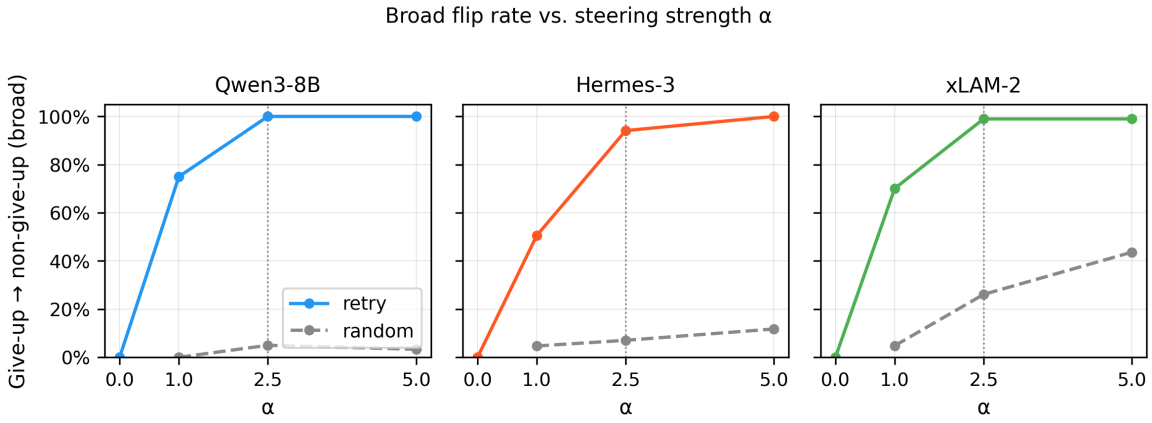


Figure 2: Broad flip rate (give-up → any non-give-up) vs. steering strength α for all three models. Solid lines: retry direction; dashed: random-direction control. Dashed vertical line marks α=2.5 (primary evaluation point).

this constraint: adding it to calculator residuals disrupts the give-up attractor at 100% broad flip rate despite the representational orthogonality. This probe-vs-causal dissociation is a structural property of the geometry, not a model-specific failure.

Tool identity is orthogonal, but the give-up→continue axis is shared. Cross-tool CKA measures the orthogonality of tool-*identity* subspaces, which on its own makes the probe-transfer failure partly definitional—orthogonal subspaces cannot support a single linear decoder. The sharper, non-definitional question is whether the *action* geometry, the give-up→continue axis itself, is shared. Within each tool we extract that axis (mean give-up minus mean non-give-up activation at the best layer) and compare it across tools (Table 7). The web_search and file_read action directions are *aligned* ($\cos = 0.62$ for Hermes-3, 0.54 for xLAM-2) even though their identity subspaces are orthogonal (CKA ≈ 0.01); the calculator action direction is orthogonal to both ($|\cos| \leq 0.18$). This converts the leave-one-tool-out pattern from a tautology into a *prediction*: probe transfer succeeds exactly between the tools whose action axes align (web_search↔file_read, LOO 50–74%) and fails for the tool whose action axis is orthogonal (calculator, 27–44%). The probe overfits to where the shared continue/give-up axis lives; the causal patch succeeds on all tools because it is applied along each trajectory's own action direction, and the subspace-matched control (Table 2) shows it is the *specific orientation*, not mere norm, that flips the behaviour.

Table 7: Pairwise cosine of per-tool give-up→continue directions at the best layer. The web_search/file_read action axis is shared; calculator's is orthogonal. Qwen3-8B is omitted (too few give-ups outside file_read for a per-tool direction).

Tool pair	Hermes-3	xLAM-2
web_search – file_read	$+0.62$	$+0.54$
calculator – file_read	-0.12	$+0.14$
calculator – web_search	-0.18	-0.13

4 Discussion

The probe-vs-causal dissociation. A striking asymmetry emerges across all three models: the causal direction forces continuation for calculator give-ups at near-100% broad flip rate, yet a probe trained on web_search and file_read trajectories cannot decode calculator labels at above-chance accuracy. The CKA

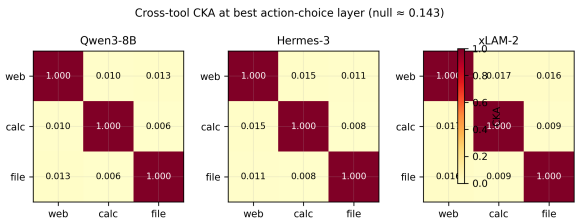


Figure 3: Cross-tool CKA at each model's best action-choice layer. All off-diagonal values ($\leq$0.017) lie far below the random null ($\approx$0.143): the three tools occupy near-orthogonal subspaces.



Figure 4: Hermes-3 vs. xLAM-2 CKA by layer. Similarity peaks at L15–16 (shared base) and drops precisely at the best action-choice layers (L28, L29).

analysis explains why: representational generalisation requires subspace alignment, which is absent for orthogonal tools, while causal disruption of the give-up attractor requires only that the injected direction have sufficient norm in the relevant subspace. Heo et al. (2024) report an analogous pattern: their instruction-following direction generalises across unseen tasks but not across out-of-distribution instruction types—representational generalisation breaks at distributional boundaries while causal effectiveness persists. The dissociation is predicted by the CKA orthogonality bound: a causal direction need not lie in the probe-trained subspace to disrupt the attractor it targets. That mechanistic prediction makes this a principled characterisation of action-choice geometry in transformers, not an experimental artefact—independent of the number of models examined.

Training depth and recovery richness. The narrow retry-same gradient (Qwen=0%, Hermes=44.7%, xLAM-2=56.7% overall; N/A, 6.3%, 34.9% for calculator) reflects each model's training distribution. Qwen3-8B's BFCL v3 scoring penalises redundant retries, making it impossible to steer toward naive looping. xLAM-2's `APIGen-MT-5k` includes transient-failure retry scenarios, producing a higher retry-same rate. The direction acts as a *continuation trigger*: it activates whatever recovery behavior the model has already learned.

Safety implication. Steering a well-trained model toward "try again" does not produce blind looping. In all models, the majority of steered trajectories produce behaviorally coherent continuation; Qwen3-8B produces *only* intelligent adaptation under steering. The asymmetry—harder to induce naive looping in better-trained models—tentatively suggests, across these three models with an informal training-depth ordering, that agent-native training may be a safety-relevant pattern. We flag this as a hypothesis motivating larger-scale controlled investigation, not a confirmed result.

Limitations. Qwen3-8B causal evaluation is limited: only 60 give-up trajectories, dominated by file_read (n=57); per-tool claims for Qwen are restricted to that tool. Intervention patches a single layer and token position; full causal mediation may involve multiple positions. The G-POS sanity check found above-chance probe accuracy at offsets t^*-1, t^*, and t^*+1, consistent with the effect not being tied to a single token slot, but a systematic multi-layer sweep remains future work. xLAM-2 uses a different tool-call format (JSON array vs. XML), which could in principle contribute to representational differences. Mock tools return deterministic, sometimes seed-invariant responses; multi-seed verification does not provide full independent replication.

5 Conclusion

A causally effective continuation direction exists in the residual stream of all three tested 8B-class LLMs: linear probes recover the 3-class action decision at 86.0–90.1% *balanced* accuracy (>52 pp above the chance

floor; $p < 10^{-4}$, Holm-Bonferroni); causal steering produces 74–95 pp absolute increase in non-give-up behaviour vs. random controls ($h > 1.86$), with a subspace-matched control confirming the specific direction—not its norm—drives the effect. Cross-tool CKA analysis reveals that the three tools occupy near-orthogonal representational subspaces (CKA ≈ 0.01 vs. null ≈ 0.14), which geometrically explains why leave-one-out probe transfer fails for the most dissimilar tool while causal effectiveness is preserved: representational generalisation and causal reach are governed by different geometric properties. Cross-model replication with xLAM-2 reveals a training-depth gradient in *recovery richness*: the direction acts as a universal continuation trigger, but the specific recovery form it elicits (retry-same vs. modify-args) tracks each model's training distribution —narrow retry-same rates of 0%, 44.7%, and 56.7% for Qwen3-8B, Hermes-3, and xLAM-2 respectively. Agent-native training enriches *how* a model recovers, not merely *whether* it does.

References

Zifeng Cheng, Jinwei Gan, Zhiwei Jiang, Cong Wang, Yafeng Yin, Xiang Luo, Yuchen Fu, and Qing Gu. Steering when necessary: Flexible steering large language models with backtracking. *arXiv preprint arXiv:2508.17621*, 2025. URL https://arxiv.org/abs/2508.17621.

Kait Healy, Bharathi Srinivasan, Visakh Madathil, and Jing Wu. Internal representations as indicators of hallucinations in agent tool selection. *arXiv preprint arXiv:2601.05214*, 2026. URL https://arxiv.org/abs/2601.05214.

Stefan Heimersheim and Neel Nanda. How to use and interpret activation patching. *arXiv preprint arXiv:2404.15255*, 2024. URL https://arxiv.org/abs/2404.15255.

Juyeon Heo, Christina Heinze-Deml, Oussama Elachqar, Kwan Ho Ryan Chan, Shirley Ren, Udhay Nallasamy, Andy Miller, and Jaya Narain. Do LLMs "know" internally when they follow instructions? *arXiv preprint arXiv:2410.14516*, 2024. URL https://arxiv.org/abs/2410.14516.

Max Klabunde, Tobias Schumacher, Markus Strohmaier, and Florian Lemmerich. Similarity of neural network models: A survey of functional and representational measures. *arXiv preprint arXiv:2305.06329*, 2023. URL https://arxiv.org/abs/2305.06329.

Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, 2019. URL https://arxiv.org/abs/1905.00414.

Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Inference-time intervention: Eliciting truthful answers from a language model. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. URL https://arxiv.org/abs/2306.03341.

Eric Todd, Millicent L. Li, Arnab Sen Sharma, Aaron Mueller, Byron C. Wallace, and David Bau. Function vectors in large language models. In *International Conference on Learning Representations (ICLR)*, 2024. URL https://arxiv.org/abs/2310.15213.

Chenlong Yin, Zeyang Sha, Shiwen Cui, Changhua Meng, and Zechao Li. The reasoning trap: How enhancing LLM reasoning amplifies tool hallucination. *arXiv preprint arXiv:2510.22977*, 2025. URL https://arxiv.org/abs/2510.22977.

Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, et al. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023. URL https://arxiv.org/abs/2310.01405.

A Validation Suite Summary

The following checks were run on all three models to validate measurement integrity.

Table 8: Selected validation checks. Most passed across all three models; two preregistered controls were not fully met and are disclosed in the text below.

Check	Qwen3-8B	Hermes-3	xLAM-2
Ambiguous label rate ($\leq$15%)	0%	0%	0%
Layer smoothness ($\leq$5 pp gap)	0.56 pp	1.67 pp	0.83 pp
PC1 class-sep. variance ($\geq$40%)	86.9%	90.6%	68.4%
Position stability (3 token positions)	3/3	3/3	3/3
Within-error-type accuracy ($\geq$50%)	3/3	3/3	3/3
Direction specificity (retry $>3\times$rand)	✓	✓	✓
Probe vs. position-only gap	+27 pp	+31 pp	+33 pp

G-SHUF resolved. The shuffled-label control (G-SHUF, target $\leq$0.40) reached 0.52, 0.41, and 0.41 under raw accuracy. Post-hoc analysis showed this was a test-set class imbalance artifact: the majority class fractions were 62%, 47%, and 56% for Qwen3-8B, Hermes-3, and xLAM-2 respectively. Under *balanced* accuracy (uniform class weighting), the true probe scores are **90.0%, 86.0%, and 90.1%** and the balanced shuffle floor sits at chance for all three models (33.8%, 33.6%, 33.5%). G-SHUF therefore passes once imbalance is accounted for. The position-only baseline (G-POS) was exceeded by Qwen3-8B (0.62) and xLAM-2 (0.56); the probe beats position-only by +27/+31/+33 pp (Table 8), establishing that residual-stream *content* encodes action choice beyond trajectory structure.

Cross-tool leave-one-out transfer. LOO probe accuracy is 83% (web_search) and 74% (file_read) when holding out web_search and file_read respectively, confirming transfer between tools with shared representational subspace. The calculator tool occupies a near-orthogonal subspace (CKA < 0.02), which geometrically bounds LOO probe transfer to that tool—a finding explained and reported as a main result in §3.4.

B Decision Log Excerpt

Full log in `logs/decisions.jsonl`. Key decision points: (1) system prompt adjusted post-pilot after Hermes-3 produced 92% give-up on initial pilot, calibrating the base-rate to a more ecologically valid distribution; (2) uniform outcome sampling adopted after resolving an internal inconsistency in the pre-registration; (3) Qwen3-8B's near-zero give-up rate ($\approx$2% of error trajectories) logged as a positive cross-model finding about agent-training quality; (4) broad metric adopted as xLAM-2 primary metric after G3 established that Qwen3-8B never produces retry-same, motivating a metric sensitive to all recovery forms; (5) cross-tool CKA analysis added after noticing the calculator LOO pattern, leading to the geometric explanation in §3.4; (6) per-tool causal table verified against raw CSVs during final review, confirming all numbers.

C Compute Attribution

Total GPU time: $\approx$5.2 h on a single NVIDIA A100 (80 GB).

Table 9: GPU time by stage.

Stage	Seconds
Trajectory gen. Hermes-3 (all seeds)	2185
Trajectory gen. Qwen3-8B (all seeds)	3053
Trajectory gen. xLAM-2 (all seeds)	721
Activation recording Hermes-3	674
Activation recording Qwen3-8B	664
Activation recording xLAM-2	74
Causal intervention Qwen3-8B	833
Causal intervention Hermes-3	2371
Causal intervention xLAM-2	2135
Total	**≈18,710**

Conference For AI Scientists 2026 – AI Involvement Checklist

Research Stage Assessment

1. **Hypothesis development.**

   Answer: **[C]**

   Iteration Effort: *[Medium]*

   Explanation: The human author identified the research target (agentic tool-error recovery) and the mechanistic framing. The AI agents proposed the specific hypotheses (H1a/b thresholds, cross-model predictions, three pre-registered fallback abstracts), generated the pre-registration document, and designed the sanity-gate suite. The human approved and committed the pre-registration before any GPU compute began, providing the critical authorial judgment checkpoint. Approximately 20–40 substantive prompt-response exchanges to reach the final pre-registration.

2. **Experimental design and implementation.**

   Answer: **[D]**

   Iteration Effort: *[High]*

   Explanation: The AI agents wrote all source code (`src/`, `scripts/`, `Makefile`), designed the probe methodology, intervention protocol, and CKA analysis pipeline. The human constrained compute resources (single GPU, CPU thread cap), approved model choices, and identified one implementation error (raw CKA with $d \gg n$ yielding an inflated null requiring a full rewrite). Hundreds of tool calls across implementation and debugging cycles.

3. **Analysis of data and interpretation of results.**

   Answer: **[D]**

   Iteration Effort: *[High]*

   Explanation: The AI agents ran all analysis scripts, computed all statistics, generated all publication-quality figures, and interpreted all results across three models. The human reviewed and validated the final numbers against raw data files before submission. Hundreds of analysis calls including multiple re-runs after error correction and CKA implementation fix.

4. **Writing.**

   Answer: **[D]**

   Iteration Effort: *[Medium]*

   Explanation: The AI agents authored all text in the paper, including the abstract, all sections, tables, captions, and both checklists. The human reviewed for factual consistency and scientific accuracy. Approximately 3–5 complete drafts across the writing process.

AI System Documentation

5. **AI systems used.**

   Description: Two frontier large language models acting in concert as an autonomous research agent. The first served as high-level research planner and decision-maker, proposing hypotheses, designing the experimental protocol, and directing the overall research strategy. The second served as the pipeline executor, writing and running all code, recording activations, training probes, running causal interventions, performing CKA analysis, verifying citations, and drafting this paper. Both models operated via a structured tool-calling loop with access to a bash shell, file read/write, and web search on a single 48 GB GPU node. One of the executor models is itself among the three agentic LLMs studied in the paper; this is disclosed in full in the camera-ready version per conference policy.

6. **Observed AI Limitations.**

   Description: Three categories of failure were observed. *Numeric hallucination:* an early draft of the paper contained a per-tool causal table labeled "broad metric" that in fact used narrow metric values, and attributed a finding to xLAM-2 that was actually shared by all models. Corrected only after a systematic verification pass against raw CSVs. *Implementation error:* the first CKA implementation used the primal $d \times d$ formula in float32; the random null was 0.72 (expected $\approx$0.93) and the sanity check failed, requiring a complete rewrite with PCA-50 and the kernel $n \times n$ formula. *Citation venue error:* Kornblith et al. was cited as "NeurIPS 2019" when the correct venue is ICML 2019. Caught only by fetching the arXiv page directly.

Conference For AI Scientists 2026 – Reproducibility and Responsibility Checklist

1. **Claims.**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: **Yes**

   Justification: All numbers in the abstract and introduction were verified against the raw result files (`causal_flip_*.csv`, `analysis_summary.json`) before the final draft. Pre-registered thresholds for H1(a) and H1(b) are stated in §2; observed values are reported in Tables 1, 2, and 3. Deviations from pre-registered secondary predictions are documented in Appendix B.

2. **Limitations.**

   Question: Does the paper discuss the limitations of the work?

   Answer: **Yes**

   Justification: Section 4 (Limitations paragraph) explicitly addresses: Qwen3-8B's small causal test set (n=60, dominated by file_read); single-layer/single-position intervention; tool-call format confound; and seed-invariant mock-tool responses preventing true independent replication.

3. **Theory Assumptions and Proofs.**

   Answer: **N/A**

   Justification: The paper is empirical; no theoretical results or proofs are claimed.

4. **Experimental Result Reproducibility.**

   Answer: **Yes**

   Justification: All hyperparameters, splits, and model identifiers are specified in §2. Mock tool responses are byte-deterministic (SHA-256 hash). The full pipeline is reproduced by `make all` from the repository root given the specified random seeds. Pre-registrations are frozen at named git commits (`0c7ea00`, `a8e84d2`).

5. **Open Access to Data and Code.**

   Answer: **No**

   Justification: Code and data are not released at submission time to preserve anonymity. All three subject models are publicly available on HuggingFace without access tokens. We commit to releasing the full pipeline code and trajectory data on acceptance.

6. **Experimental Setting / Details.**

   Answer: **Yes**

   Justification: §2 specifies model identifiers, layer counts, hidden dimensions, label rubric, probe hyperparameters (ℓ_2, 5-fold CV), intervention formula, α sweep values, and all three null controls.

7. **Experiment Statistical Significance.**

   Answer: **Yes**

   Justification: Probe accuracy is tested by permutation test (10k shuffles) with Holm-Bonferroni correction (family-wise α=0.05). Causal flip rates use cluster bootstrap by base-prompt (10k resamples, percentile bootstrap CI). Wilson 95% confidence intervals are shown as subscripts for all flip-rate point estimates in Tables 2, 3, and 5. Cohen's h effect sizes are reported for all causal comparisons.

8. **Experiments Compute Resources.**

   Answer: **Yes**

   Justification: Appendix C lists GPU seconds per stage. All experiments ran on a single NVIDIA A100 (80 GB) using `bfloat16` precision; total GPU time $\approx$5.2 h. CPU analysis used $\leq$16 threads (OMP/MKL/OpenBLAS cap enforced).

9. **Research Ethics.**

   Answer: **Yes**

   Justification: The study uses only publicly available, open-weight language models and synthetic mock-tool data; no human subjects, personal data, or proprietary systems are involved. The AI authorship structure follows CAISc 2026 policy and is fully documented in the AI Involvement Checklist above.

10. **Broader Impacts.**

    Answer: **Yes**

    Justification: §4 (Safety implication paragraph) discusses the positive implication that agent-native training makes naive looping *harder* to induce via residual-stream patching. A potential negative impact is that the causal direction could in principle be used to force over-persistent behavior in deployed agents; we note that the direction must be applied by an entity with full model-weight access, limiting practical misuse.